

1. (i) All divisors of $n = 2^6 \cdot 3^4 \cdot 5^3$ must have the expression $2^{\alpha_1} \cdot 3^{\alpha_2} \cdot 5^{\alpha_3}$

where $\alpha_1 \in \{0, 1, \dots, 6\}$, $\alpha_2 \in \{0, 1, \dots, 4\}$, $\alpha_3 \in \{0, 1, \dots, 3\}$.

Hence, there are $7 \cdot 5 \cdot 4 = 140$ positive divisors.

(ii) perfect square $\Leftrightarrow \alpha_1, \alpha_2, \alpha_3$ are even numbers.

$$\text{Total number} = 4 \cdot 3 \cdot 2 = 24.$$

(iii) $\alpha_1 \geq 2$ & $\alpha_2 \geq 1$. $5 \cdot 4 \cdot 4 = 80$.

(iv) $30 = 2 \cdot 3 \cdot 5 \Rightarrow \alpha_1 = \alpha_2 = \alpha_3 = 0$. only 1.

2. (i) $10 = 2 \cdot 5$. $\phi(10) = 10 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 4$. $\gcd(3, 10) = 1 \Rightarrow 3^4 \equiv 1 \pmod{10}$.

$$\Rightarrow 3^{100} = (3^4)^{25} \equiv 1 \pmod{10}.$$

(ii) $40 = 2^3 \cdot 5$. $\phi(40) = 40 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 16$. $\gcd(7, 40) = 1 \Rightarrow 7^{16} \equiv 1 \pmod{40}$.

$$\Rightarrow 7^{222} = (7^{16})^{14} \cdot 7^{-2} \equiv 7^{-1} \pmod{40} \equiv 9 \pmod{40}$$

(iii) $6^{50} \equiv 0 \pmod{3}$

$$6^{50} \equiv 1 \pmod{5} \Rightarrow 6^{50} \equiv 6 \pmod{15}$$

(iv) $8^{123} \equiv 0 \pmod{4}$

$$8^{123} \equiv 3^{123} \equiv 9^{61} \cdot 3 \equiv -3 \pmod{5} \Rightarrow 8^{123} \equiv 12 \pmod{20}.$$

3. (i) $6x \equiv 15 \pmod{21} \Leftrightarrow 2x \equiv 5 \pmod{7} \Leftrightarrow x \equiv 6 \pmod{7}$

(ii) $\gcd(8, 20) = 4$. $4 \nmid 7$. No solution.

(iii) $3 \rightarrow 10 \rightarrow 17$. $x \equiv 17 \pmod{28}$

(iv) $x \equiv 3 \pmod{6} \Rightarrow x \equiv 1 \pmod{2}$, contradicting with $x \equiv 2 \pmod{4}$. No solution.
 $\uparrow$ odd $\uparrow$ even.

4. (i) $\gcd(a, b) = 1 \Rightarrow ax' + by' = 1$ for some $x, y \in \mathbb{Z}$.

$$\Rightarrow ax' \equiv 1 \pmod{b} \Rightarrow ax'n \equiv n \pmod{b}$$

Take $x = x'n$.

(ii) Assume $a > b$. Let $x \leq b-1$ be the minimum positive solution to

$$ax \equiv n \pmod{b}.$$

If $ax \leq n$, taking $y = (n - ax)/b$ gives the desired solution.

If $ax > n$, then $x = b-1$ as $n \geq (a-1)(b-1) = a(b-1) - (b-1) > a(b-1) - b > a(b-2)$.

$ax \equiv n \pmod{b}$ & $ax > n$ imply that $ax - b \geq n$.

Hence $n \leq a(b-1) - b < (a-1)(b-1)$. Contradiction.